

# Chengle Fan

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## EDUCATION

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### Southern University of Science and Technology

Shenzhen, China

*B.Eng. in Optoelectronic Information Science and Engineering*

*Expected June 2027*

- GPA: 3.85/4.00; Weighted Average: 92.45/100; Major Rank: 7/46.
- Selected coursework: Fundamentals of Optics, Engineering Electromagnetics, Solid-State Electronics, Introduction to Semiconductor Optics, Liquid Crystal Optoelectronics, Optical System Design Experiment.

## RESEARCH INTERESTS

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Artificial photonic structures and nanophotonic systems, including photonic crystals, metamaterials, and metasurfaces, for controlling light propagation and engineering light-matter interactions. Specific interests include topological and time-varying phenomena, nonreciprocity, and reconfigurable photonic functionalities.

## MANUSCRIPT IN PREPARATION

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**Chengle Fan**, Xiang Xi, and Zhen Gao, “Observation of pseudospin-locked chiral edge states in a gyromagnetic photonic crystal,” *Manuscript in preparation*.

## RESEARCH EXPERIENCE

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### Undergraduate Research Assistant

Feb. 2025–Present

*Topological Physics Research Group, Department of EEE, SUSTech*

*Supervisor: Prof. Zhen Gao*

*Pseudospin-Locked Chiral Edge States in Gyromagnetic Photonic Crystals*

- Led full-wave modeling and sample design for a microwave gyromagnetic photonic crystal supporting pseudospin-locked chiral edge transport under broken time-reversal symmetry.
- Identified QSH-like, QAH-like, and trivial phases through bulk and projected band structures,  $p$ - $d$  band inversion, pseudospin-resolved eigenfields, and spin Chern numbers.
- Designed measurement configurations and participated in VNA transmission and near-field mapping; measurements showed approximately 20 dB directional transmission contrast within the bandgap and pseudospin-selective excitation of the chiral edge mode.
- Designed a pseudospin multiplexer/demultiplexer and a low-crosstalk waveguide crossing based on the coexistence of multiple topological phases.

### *Gauge-Field-Induced Duality and Hofstadter Photonic Lattices*

- Developed tight-binding and full-wave models of gauge-flux-engineered dual and self-dual photonic crystals, verifying duality-related spectral equivalence and self-duality-induced degeneracies.
- Engineered rotation-dependent effective couplings of higher-order photonic modes to implement spatially modulated hopping amplitudes and investigate Hofstadter-butterfly-like spectra.
- Built finite-array simulations and MATLAB workflows for spectral reconstruction and Fourier analysis, and explored how synthetic gauge flux, coupling modulation, and duality affect the resulting fractal band structure.

### *Cavity-Mediated Topological Transport in Metallic Dipole Arrays*

- Conduct microwave transmission and near-field measurements and process S-parameter and field data for an ongoing collaboration between Prof. Gao’s and Prof. Mengyao Li’s groups on cavity-tunable topological transport in all-metal dipole arrays; reconstruct dispersions by FFT and compare them with simulated bands.

## Visiting Research Student

*School of Physical and Mathematical Sciences, NTU*

Jan. 2026–Apr. 2026

*Supervisor: Prof. Baile Zhang*

*Terahertz Chiral Valley Edge States in Magneto-Optic Photonic Crystals*

- Designed and numerically validated a fabrication-compatible terahertz Chern–valley photonic-crystal interface supporting chiral, valley-selective edge modes.
- Built 2D and 3D COMSOL bulk-band and supercell models to identify overlapping topological gaps and interface-localized modes.
- Evaluated magnetic-bias configurations, fabrication tolerances, domain alignment, and finite air-layer effects, and re-optimized the structure to provide parameters for subsequent fabrication and experiments by NTU collaborators.

## RESEARCH GRANT

### SUSTech Undergraduate Research Training Program

*Project Leader*

Shenzhen, China

*June 2026–June 2027*

- Secured a competitive university research grant of CNY 10,000 for “Design of 3D Photonic Topological Insulators at Telecommunication Wavelengths.”

## TECHNICAL SKILLS AND RESEARCH METHODS

- **Computational electromagnetics:** COMSOL Multiphysics and MATLAB for eigenmode and frequency-domain modeling of photonic crystals, metamaterials, and metasurfaces, including bulk/projected bands, eigenfield reconstruction, finite structures, and dispersion analysis.
- **Band topology and numerical modeling:** Tight-binding modeling; Berry phase and curvature, Chern number, and Wilson-loop calculations; synthetic gauge fields, wave-packet dynamics, and FFT-based dispersion reconstruction.
- **Time-varying and spatiotemporal photonics (ongoing research training):** Undertaking structured training in Prof. Gao’s group under the guidance of a postdoctoral researcher, focusing on MATLAB-based PWE/Floquet–Bloch band calculations, eigenfield reconstruction, and FDTD simulations for photonic time crystals and space–time-periodic media.
- **Experimental characterization:** VNA-based microwave transmission spectroscopy, near-field mapping, chiral-source excitation, S-parameter acquisition and analysis, and spatial-Fourier reconstruction of dispersions; experience with optical alignment, CCD imaging, polarization microscopy, liquid-crystal cell assembly, and basic electro-optic measurements.
- **Programming and software:** MATLAB for numerical eigensolvers, time-domain simulations, data processing, visualization, and COMSOL LiveLink automation; basic experience with C and Meep FDTD; familiarity with Zemax and TracePro.
- **Languages:** Mandarin Chinese (native); English (TOEFL iBT 104).

## HONORS AND AWARDS

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| • SUSTech Outstanding Student Leader                         | 2025       |
| • SUSTech Outstanding Student                                | 2024, 2025 |
| • SUSTech Excellence Scholarship, Second Class / Third Class | 2024, 2025 |

## LEADERSHIP AND ACTIVITIES

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| • <b>Deputy Head, Activities Department</b> , Student Union, Department of EEE, SUSTech | 2024–2025 |
| • <b>Vice President</b> , SUSTech Orienteering Club                                     | 2024–2025 |